WORKED SOLUTION

STEP III 2006, Question 8

Pure • Axiomatic Differentiation

STEP III 2006, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q8

The key idea

The four rules are exactly the properties of differentiation you use every day: linearity and the product rule, plus the single fact that x "differentiates to" 1. Every polynomial is built from the number 1 and the letter x by multiplying and adding, so if we know what $\triangle$ does to 1 and to x, the rules leave it no freedom anywhere else. The product rule applied to $x \cdot x^n$ gives an induction on powers, and linearity then finishes the job.

Solution

Throughout, "rule (i)" to "rule (iv)" refer to the four defining rules, in the order given.

Constants

First take the constant polynomial 1. Since $1 = 1 \cdot 1$, rule (iv) with $f(x) = g(x) = 1$ gives

$$\triangle 1 = \triangle(1 \cdot 1) = 1 \cdot \triangle 1 + 1 \cdot \triangle 1 = 2\triangle 1.$$

Subtracting $\triangle 1$ from both sides gives $\triangle 1 = 0$ (the zero polynomial).

Ask yourself Why not just say "$\triangle$ of a constant is zero, like a derivative"?

Because we don't yet know that $\triangle$ *is* a derivative; that is what the last part asks us to prove. All we may use are the four rules, and none of them mentions constants directly. The trick is that 1 is its own square, so the product rule forces $\triangle 1$ to equal twice itself.

Now let $f(x) = \lambda$ be any constant. Then $\lambda = \lambda \cdot 1$, so by rule (iii)

$$\triangle\lambda = \triangle(\lambda \cdot 1) = \lambda\triangle 1 = \lambda \cdot 0 = 0.$$

So $\triangle f(x) = 0$ for every polynomial of degree zero, as required.

The square and the cube

Writing $x^2 = x \cdot x$ and using rule (iv), then rule (i),

$$\triangle x^2 = x\,\triangle x + x\,\triangle x = x \cdot 1 + x \cdot 1 = 2x.$$

Writing $x^3 = x \cdot x^2$ and using rule (iv), then the result just found and rule (i),

$$\triangle x^3 = x\,\triangle x^2 + x^2\,\triangle x = x \cdot 2x + x^2 \cdot 1 = 3x^2.$$

The general polynomial

Step 1: powers of x. We prove by induction that

$$\triangle x^n = n\,x^{n-1} \qquad \text{for every integer } n \geq 1.$$

For $n = 1$ this says $\triangle x = 1$, which is rule (i). Suppose it holds for some $n = k \geq 1$. Since $x^{k+1} = x \cdot x^k$, rule (iv) gives

$$\begin{aligned} \triangle x^{k+1} &= x\,\triangle x^k + x^k\,\triangle x \\ &= x \cdot k\,x^{k-1} + x^k \cdot 1 \\ &= (k+1)\,x^k, \end{aligned}$$

using the induction hypothesis and rule (i) in the second line. So the result holds for $n = k+1$, and by induction for all $n \geq 1$. For $n = 0$, $x^0 = 1$ and we showed $\triangle 1 = 0$, which agrees with $\dfrac{\mathrm{d}}{\mathrm{d}x}1 = 0$. So for every $n \geq 0$,

$$\triangle x^n = \frac{\mathrm{d}}{\mathrm{d}x}\,x^n.$$

Step 2: sums of many terms. Rule (ii) is stated for two polynomials, but it extends to any finite number. If $\triangle(f_1 + \cdots + f_m) = \triangle f_1 + \cdots + \triangle f_m$, then applying rule (ii) to the two polynomials $f_1 + \cdots + f_m$ and f_{m+1} gives

$$\triangle(f_1 + \cdots + f_{m+1}) = \triangle(f_1 + \cdots + f_m) + \triangle f_{m+1} = \triangle f_1 + \cdots + \triangle f_{m+1},$$

so by induction on m it holds for any number of terms.

Step 3: putting it together. Any polynomial can be written $h(x) = \sum_{k=0}^{n} a_k x^k$ with constants a_k. Then

$$\begin{aligned} \triangle h(x) &= \sum_{k=0}^{n} \triangle\left(a_k x^k\right) && \text{by rule (ii), extended in Step 2} \\ &= \sum_{k=0}^{n} a_k\,\triangle x^k && \text{by rule (iii)} \\ &= \sum_{k=1}^{n} k\,a_k\,x^{k-1} && \text{by Step 1 (the } k = 0 \text{ term is 0)} \\ &= \frac{\mathrm{d}h(x)}{\mathrm{d}x}, \end{aligned}$$

the last line being the usual derivative of a polynomial. This holds for every polynomial $h(x)$, so $\triangle h(x) \equiv \dfrac{\mathrm{d}h(x)}{\mathrm{d}x}$, as required.

Checks

It is reassuring that ordinary differentiation obeys all four rules, so the rules aren't contradictory and $\mathrm{d}/\mathrm{d}x$ really is an operation of the kind described. You can also test the product rule against the formula: for $f = x^2$ and $g = x^3$, rule (iv) gives $x^2 \cdot 3x^2 + x^3 \cdot 2x = 5x^4$, which is $\triangle x^5$. In a computer we built $\triangle$ using only the rules (as in Steps 1 to 3) and compared it with the derivative, exactly, for 2000 random polynomials of degree up to 9; for instance $\triangle\left(\frac{1}{2}x^4 - 3x + 7\right) = 2x^3 - 3$, which also matches a numerical difference quotient.

Common slips

- Asserting $\triangle(\text{constant}) = 0$ "because it's a derivative". That assumes the final answer; it has to be derived from the rules.
- Using rule (ii) for a long sum without comment. It's stated for two polynomials, so a sentence of induction is needed for more terms.
- Not saying which rule is used where: the question explicitly asks for this.

Beyond STEP

An operation obeying linearity and the product rule is called a *derivation*, and the argument here shows that on polynomials a derivation is completely fixed by its value on x. If instead $\triangle x = p(x)$ for some polynomial p, the same proof gives $\triangle h = p(x)\,h'(x)$. The idea that differentiation can be defined algebraically, with no limits at all, is central in abstract algebra and in differential geometry, where vector fields are exactly the derivations on functions. Allenby's *Rings, Fields and Groups* is a good place to meet polynomial rings treated algebraically.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.